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Low Volume Sampler LVS/LVS6-RV

SMALL FILTER DEVICE (Kleinfiltergerät – KFG)

Features

- Rugged, light-weight construction for outdoor use (**stainless steel**), small set-up area
- Self-explanatory menu-guided operation by means of 3 keys:
Start and stop by hand
Automatic operation
Operations in intervals
- Data Storage on **USB-stick (data logger function)**
- **Bavarian-Hessian protocol**
- Protection of stored data against power failure, real-time clock
- Device will be automatically heated (winter operation) and ventilated
- Controlling of **operating-m³/h** (ambient air conditions) by orifice
- Impactor inlets with exchangeable jets (8 pieces) for **PM10 – PM4,0 – PM2,5 – PM1,0**
- Inlets for **TSP, PU foam** (with and without **ozone denuder / CEN EN 15549**) and **bioaerosols**
- External set-up of inlets
- Use of filters with diameter of **47 mm** and **50 mm**



LVS

according to VDI 2463-7

**PM10 / PM2,5 Standard Reference
Sampler according to CEN EN 12341:2014**

Description

The LVS sampler LVS6-RV can be operated at temperatures of approx. $-30\text{ }^{\circ}\text{C}$ up to more than $50\text{ }^{\circ}\text{C}$ under all weather conditions. The sampler can also be used indoors..

The **flowrates** are controlled in terms of operating $\text{--m}^3/\text{h}$ in compliance with basic physical principles by means of a temperature- and pressure-compensated orifice according to Bernoulli's law and by conversion to ambient conditions according to Boyle-Mariottes's law.

The flow rates as well as the sampled volumes are displayed and recorded (**USB-Stick**) in terms of **operating- m^3/h** resp. **operating- m^3** (related to ambient conditions, each) and additionally recorded in terms of **standard- m^3/h** resp. **standard- m^3** (related to $0\text{ }^{\circ}\text{C}$ and 1013 mbar , each).

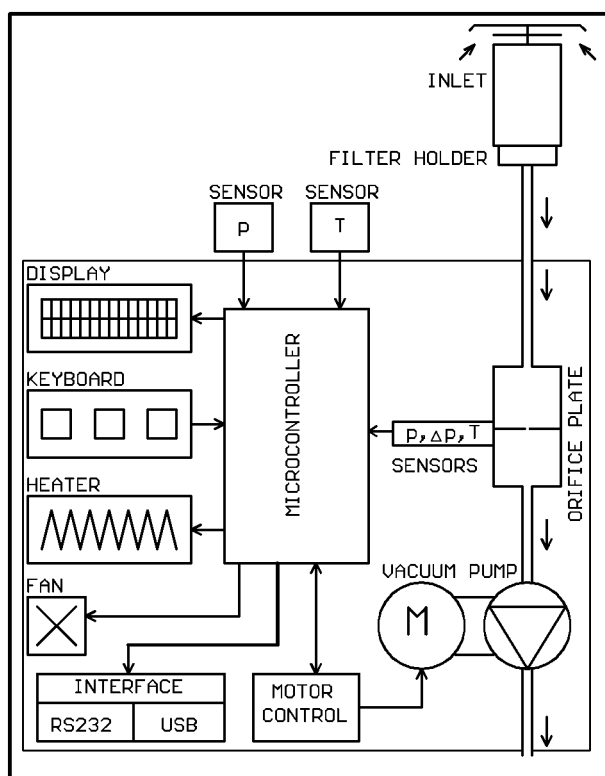
Further, all **relevant data** is displayed and can be recorded on the **USB stick with data logger function**. Direct data transfer **RS232** to a PC is possible e. g. by means of the **Bavarian-Hessian protocol**.

At a pressure drop across the currently sampled filter of $> 300\text{ mbar}$ the sampler shuts down.

In case of a power failure, all data stored in the micro-controller and in the system's memory will be safe for several years thanks to a built-in high-capacity battery.

The **filter cassettes**, which are placed in the sampling inlet directly downstream the impactor, consist of **POM**.

The sampling head can be set up externally, e.g. directly at kerbsides or in living rooms by using a hose connection to the sampler placed separately in a certain distance.



BLOCK DIAGRAM LVS

The flow rates as well as the temperature and pressure sensors can be easily recalibrated by means of the 3 front keys.

Inlets

For ambient air, indoor and workplace measurements

- Measurement of PM10 and PM2,5 (EN 12341)
- Measurement of PM4,0 and PM1,0
- Measurement of TSP (VDI 2463-8)
- Measurement of heavy metals (VDI 2267 and EN14902))
- Measurement of PCBs (VDI 2464-1)
- Measurement of soot (EC/OC) (VDI 2465 and CEN/TR 16243)
- Measurement of dioxins and furans (VDI 3498-2 and ISO 16000-13)
- Measurement of bioaerosols (VDI 4252-2)
- Measurement of PAHs, PCDD, PCDF, PCB and house dust (VDI 4300, ISO 16000-13 and Directive 2004/107/EC)
- Measurement of lindane/PCP, house dust etc. (VDI 4301)
- Measurement of BaP (Scrubber/EN 15 549)

The dust collected on the filters can be also analysed on ions (sulphate, nitrate etc.) as well as radioactivity.

Technical Data

Flow rates

Controlled 2,3-2,7-3,0 m^3/h
Deviation from the set point: $< 2\%$

Sampling time

minimum 1 h – maximum 168 h

Breaks (interval mode)

minimum 1 h – maximum 168 h

Power supply

230 V, 50/60 Hz

Consumption

approx. 300 VA

Filter diameter

47 mm and 50 mm

Diameter of sampled filter surface

approx. 40 mm

Dimensions

Width 310 mm – Height 480 mm – Depth 250 mm

Weight

approx. 23 kg

Noise level according to DIN 2058

$<< 35\text{ dBA}$